Assessing the Impact of Earthquakes on Food Security in Post-Disaster Nepal: Evidences from the 2015 Earthquake

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Abstract

Nepal has always been a country that has been prone to earthquakes due to tectonic plates- the Indo-Australian and Asian plates. Nepal is also primarily an agrarian economy with 23.95% of agricultural contribution in the country's GDP in fiscal year 2021/22 with the involvement of about 21.5% of the population. The earthquakes cause significant changes in the labor, income, land quality and availability of the affected area. This research paper analyzes the impacts of earthquake in food security of households in Nepal. We have drawn the data from the Household Risk and Vulnerability Survey conducted by the World Bank from 2015/16 to 2017/18. This research explores the socio-economic condition of households after the 2015 earthquake. This links with how these variables impact the food security index of the given household. The statistical model proposed in the study will show how the food security index of that particular household fluctuates with the change in earthquake intensity.

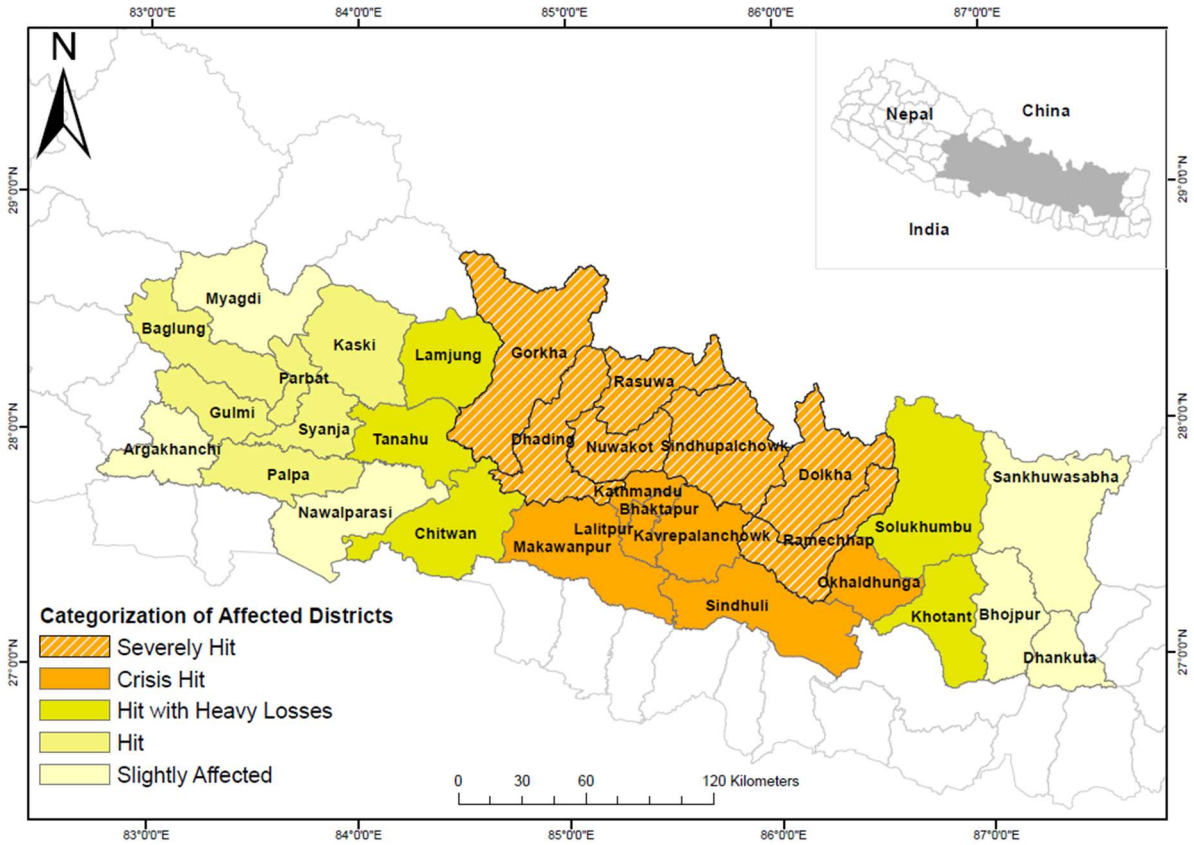
Keywords: Agriculture, Agricultural Yield, Earthquake, Earthquake Loss, Food Security

1. Introduction

Nepal is a low-income developing nation where agriculture accounts for 21.9% of the population's employment and contributes 23.95% of the GDP. According to estimates, the agriculture sector grew by 2.3% in 2022 compared to 2.85% the year before (Nepal Rastra Bank, 2022; National Statistics Office, 2018). Agriculture is the primary occupation of vast majority of economically active poorest households in Nepal. With the rapid population growth and increasing urbanization, Nepal has seen a steep increase in demand for agricultural products. Over 80% of the farmers are either landless/ near landless farmers who collectively only account for 47% of the total land available. While the remaining 20% own the remaining 53% of the land, and engage in commercial agriculture (Ministry of Agriculture Development, 2016). About three-fourth of farmers produce crops for home consumption (Samridhi, 2011).

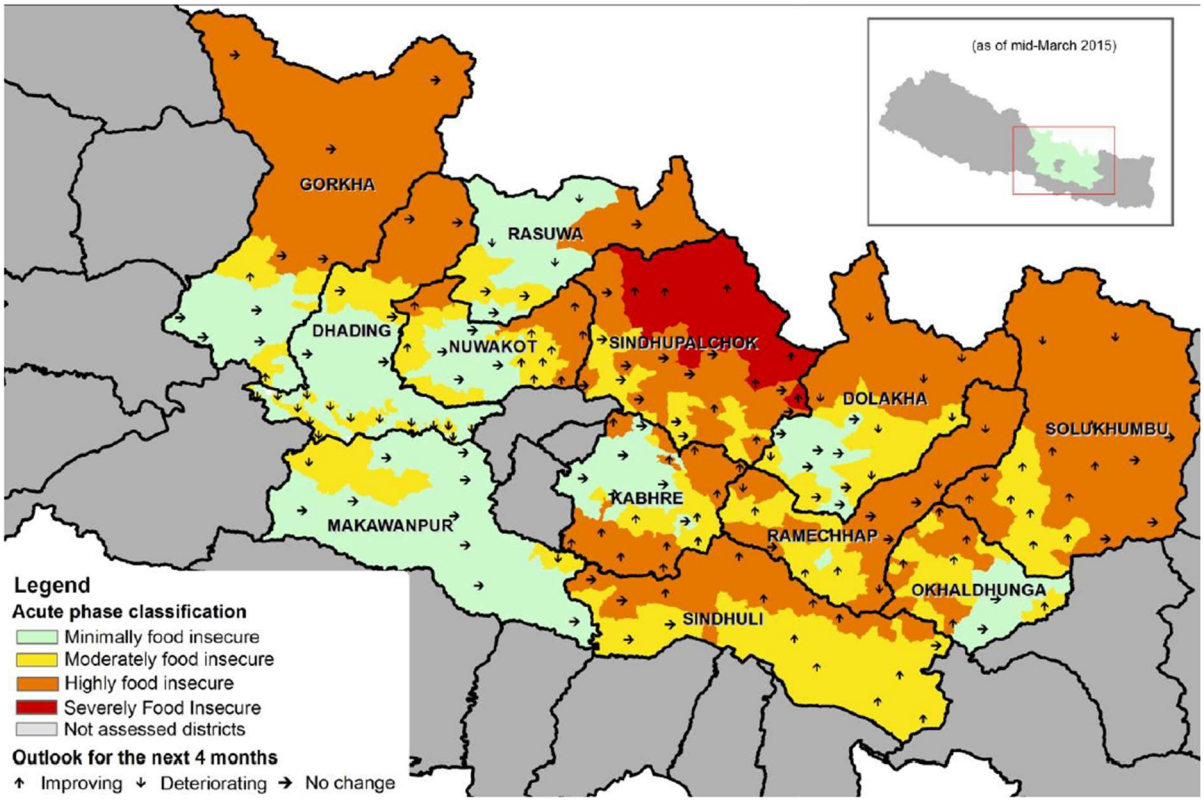
Nepal is situated between the Eurasian and Indian plates, which are what gave origin to the Himalayas. Nepal has seen 1248 earthquakes with a magnitude greater than 4.0 on the Richter scale between 1994 and 2023. As a result, 42 earthquakes with an intensity of more than 4.0 occur in Nepal annually on average. The most recent one, which had an epicenter in the Jajarkot district with an intensity of 6.4, produced roughly 483 aftershocks. (Nepal Earthquake Monitoring and Research Centre, 2023).

One type of natural calamity that can still not be foreseen is earthquakes. (United States Geological Survey, n.d.). Nepal, due to its geophysical structure, unfavorable climatic conditions and tectonic plate movements, ranks eleventh most earthquake-prone countries in the world. The 7.8 magnitude earthquake, which struck Nepal on April 25, 2015, caused NPR 706 billion worth of damage. In particular, around a million small farming households in the 24 hardest-hit areas had negative effects in the agriculture sector (Wendelbo, China, Dekeyser, Taccetti, Mori, Aggarwal, Alam, Savoldi, Zielonka, 2016). The earthquake induced 4000 landslides, and caused about 9000 deaths, more than 23000 injured and over 2 million people displaced (Basnyat, Tabin, Nutt, & Farmer, 2015; Government of Nepal National Planning Commission, 2015; Kargel et al., 2016). Furthermore, in the 14 most affected districts, the houses lost their stored grain and seeds, livestocks, along with that roads and markets were destroyed which play a primary role in meeting the food demands (Webb, West Jr, O'Hara, 2015). The 2015 earthquakes wiped out 25% of that year's GDP (WFP, 2016).



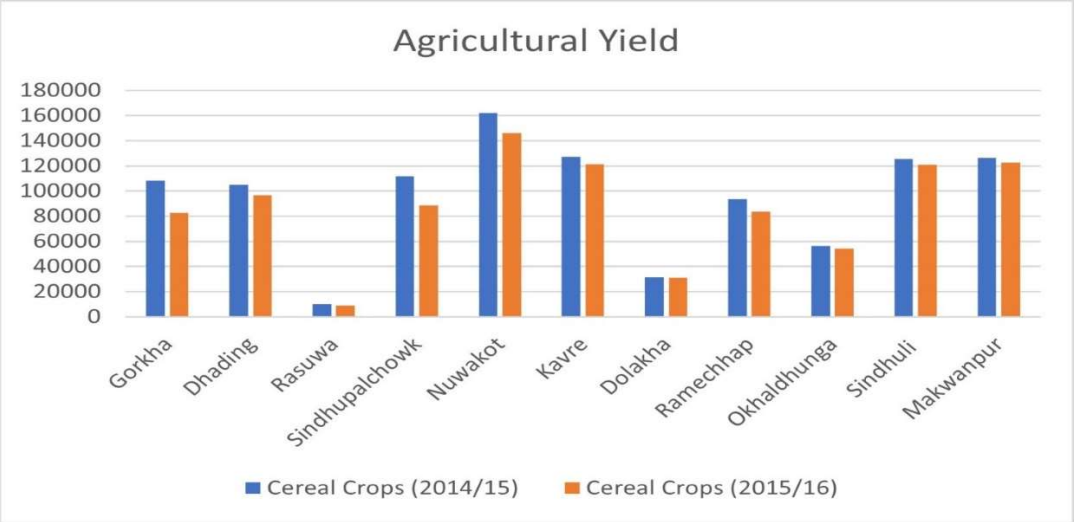
Source: Post Disaster Risk Assessment, NPC 2015.

Food Security is defined when all people, at all times, have physical and economic access to sufficient safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life (World Food Summit, 1996). Food security, as defined by the United Nations' Committee on World Food Security, means that all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their food preferences and dietary needs for an active and healthy life (1974). According to the World Bank, there are four dimensions to food security- availability of food, accessibility, utilization and stability. The supply side of agriculture, which includes food product commerce, stockpiles, and output levels, is referred to as food availability. The ability of a person to obtain food products is referred to as accessibility. Put another way, it highlights a person's physical and financial access to food, as well as their ability to afford the essential dietary items. It places more emphasis on income, spending, markets, and prices than it does on food supply in order to achieve food security goals. The term food utilization describes the variation in food consumption based on the nutrition scale. It is concerned with an individual's dietary intake of nutrients, as these can be essential to maintaining one's health. And last, stability is the ability of an individual to consistently eat enough food. Even though someone has enough food for the day, they may still be deemed food vulnerable if they don't regularly have access to food, which causes their nutritional status to decline (n.d.).



Source: Nepal Food Security Bulletin 2015, Issue 45

Agriculture was the dominant livelihood activity in the affected areas, with around 70% of households engaging in crop production and almost all households owning a land (GoN, NFSMS, WFP, 2015). About 135,200 tonnes of foodstuff, 16,399 large livestock, 36,819 small livestock, and 60,762 poultry animals have been lost. The Post Disaster Need Assessment (PDNA) from National Planning Commission (2015) and Food and Agriculture Organization study (2015) suggest that rice and millet grains harvested in November and stored in houses as stocks were destroyed.



Source: Compiled from MoFA 2015 and 2016.

Nepal, in 2018, enshrined food as a fundamental right of every step as Right to Food Act (Nepal Law Commission, 2018). Nepal government also introduced The Agricultural Development Strategy (ADS), 2015-2035, which places a strong priority on competitiveness, enhancing lives, access to food and nutrition in the commercial agriculture sector of the country (FAO, 2015).

2. Objectives of the Study

This paper aims to investigate how the 2015 Nepal earthquake affected a household's food security, taking into account factors including exposure to the earthquake, the value of education, domestic production, road connectivity, health expenses, total income, and losses caused by the earthquake. With an emphasis on the unique difficulties faced by households in Nepal, this study seeks to shed light on the complex dynamics affecting food security in post-disaster contexts. It also offers recommendations for more potent policy interventions to improve food security in vulnerable areas. The goal of this research is to provide a comprehensive understanding of the factors that affect food security in the setting under study.

3. Literature Review

There are minimal studies explaining the relationship between food security and earthquake, in the context of Nepal. However, the ones done have given empirical evidence that earthquake has caused decline in agricultural productivity. The studies examined the relationship between earthquake, rainfall and food security. Randell, Jiang, Liang, Murtugudde and Sapkota (2021) have shown that there is a negative interrelationship between earthquake and rainfall and food security, with the introduction of slope, a variable that'd give steepness of the cluster, where they'd be more susceptible to landslides. It showed that households, who experienced moderate to severe shaking, in steeper slope or in hilly regions were more susceptible to food insecurity than ones in less steep regions.

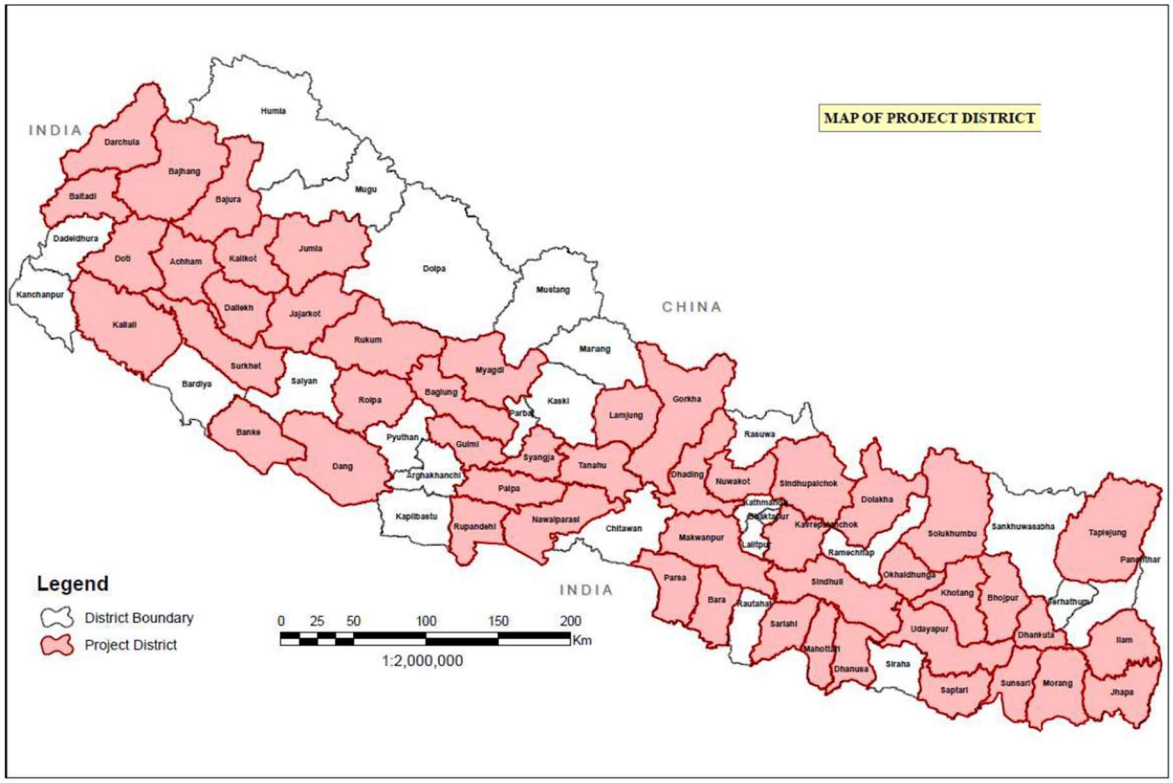
Epstein, DiCarlo, Marsh, Adhikari, Paudel, Ray and Maren (2018) noted that the damages to fields and terraces impeded subsequent grain cultivation. There were disruptions in canal system which resulted in less reliable water supply for rice cultivation. In the monsoon farming season, a year later the earthquake, several farmers changed their planting regimes. There was large decrease in total acreage planted because of damaged terraces and canals, scarcity and high labour costs and reduced seed stocks.

Thorne-Lyman, K.C., Manohar, Shrestha, Nonyane, Neupane, Bhandari, Kelmm, Webb and West Jr. (2018) reported that household food insecurity significantly decreased from 2014 to 2016. Dietary quality had also improved from the prior years. Despite the shocks the households experienced, they found that most indicators related to nutrition were stable, if not better compared to the pre-earthquake condition. They suggested that relief and rehabilitation efforts had been effective in stabilizing the nutritional situation in earthquake affected households.

4. Research Methodology

To understand the effects of earthquake in food security, we used the micro data collected in phase 1 of the 2016 Household Risk and Vulnerability Survey (HRVS) conducted by the World Bank. The HRVS uses a stratified clustered sampling design to select representative household samples at the national, provincial and rural/urban levels. 50 out of 77 districts were divided into rural and urban areas for a total of 400 sampling units and approximately 15 households in each sampling unit. Data were collected between June 2016 and August 2016 among 6000 households. Our analytic sample consists of 5998 households with non-missing values on variables of interest.

Map of 50 Survey Districts

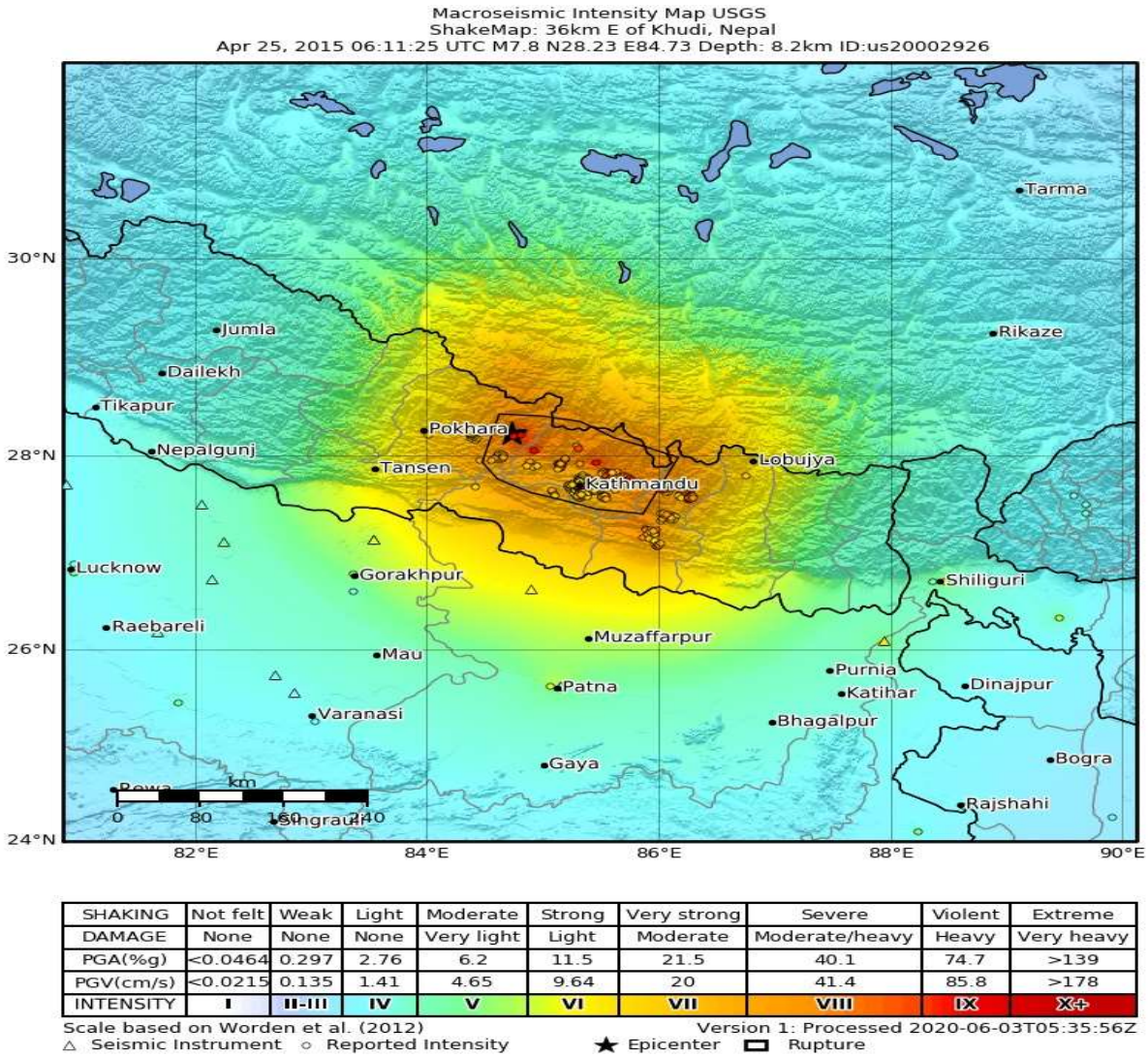


Source: HRVS, Wave 1 Report 2016, World Bank.

We created a measure of household food security based on question asked to the heads of household where they were asked how often they worried about not having enough food in the past 12 months. If they answered yes, it was quantified as 0 and 1 if not. In other words, 1 represents a food secure household and 0 represents food insecure household. Among the analytical sample, 71% of the household were food secure and the remaining 29% were food insecure.

We included a measure of earthquake ground shaking intensity developed by the US Geological Survey (USGS) Earthquake Hazards Program (USGS, 2015). The USGS's ShakeMap intensity scale ranges from one (no damages, not felt) to ten (extreme, heavy damage). The maximum intensity of the Gorkha earthquake was eight (severe, moderate/heavy damage), with most areas experiencing

intensities ranging from five (moderate, very light damage) to seven (very strong, moderate damage). Fig 5 represents the ground shaking intensity data for the Gorkha earthquake.



Source: Macroseismic Intensity Map, USGS, 2015

As represented in figure 1, the severely hit 13 districts experienced most severe impacts in terms of death, injuries and damages to infrastructures. Therefore, most of the aid packages and humanitarian assistance were concentrated in those districts (MoHA, 2015). In the sample, 16% of the households experienced moderate shaking, 41% experienced strong to severe shaking, and roughly 43% of the households experienced weak to light shaking.

The statistical model used to understand the relationship between Food Security and Earthquake intensity in Nepal is given below,

$$FSI_d = \beta_0 + \beta_1 EQ_t + \beta_2 PI + \beta_3 L + \beta_4 RL + \beta_5 HPU + \beta_6 H + \beta_7 HP + \beta_8 GP + \beta_9 EL + \beta_{10} I + \epsilon$$

Where,

FSI_d = Food Security Index of the household

EQ_t = Earthquake Intensity

PI = Personal Income

L = Literacy

RL = Road Linkage

H = Health Expenses

I = Illness

HP = Home Production Value

HPU = Home Purchase Value

GP = Government Subsidy

EL = Earthquake Loss

ϵ = Error

The sum of the household's members' incomes was used to determine the Personal Income (PI). Every family member was questioned about their monthly income. Similarly, we utilized 12 years of schooling for secondary education, 16 for undergraduate education, 18 for graduate education, and 20 for doctorate education for calculating Literacy (L), measured in years. The Road Linkage (RL) indicates the distance in kilometers between the dwelling and a motorable road. When asked if there had been any Illnesses (I) in the home in the previous year, the heads of the households were queried. If they said "yes," we represented that as 1, if not 0. The monetary worth of the goods the household created for consumption is represented by the Home Production Value (HP), and the monetary value of the food the household purchased for consumption during a given week is represented by the Home Purchase Value (HPU). The estimated monetary value of the household's lost assets in the earthquake is known as the Earthquake Loss (EL). Government Policies (GP) were computed in binary form: a family was assigned a number of 1 if a member received subsidies or was enrolled in social security; otherwise, the household was assigned a number of 0.

5. Results and Discussions

Table 1: Regression coefficients of the given model.

Variables	Estimate	Standard Error	Statistic	P-value	
Intercept	0.419	0.030	13.949	1.52E-43	***
Earthquake Intensity	0.069	0.005	12.461	3.32E-35	***
Illness	-0.057	0.011	-4.978	6.57E-07	***
Government Subsidy	-0.055	0.012	-4.437	9.26E-06	***
Literacy	0.002	0.000	5.537	3.20E-08	***

Road Linkage	-0.001	0.000	-4.581	4.71E-06	***
Health Expenses	2.21E-07	1.82E-07	1.212	0.225	
Home Production	4.90E-05	7.58E-06	6.460	1.13E-10	***
Home Purchase	-3.82E-05	6.36E-06	-6.012	1.93E-09	***
Earthquake Loss	-9.84E-08	3.12E-08	-3.155	0.001	**
Total Wages	-1.94E-07	2.56E-07	-0.755	0.449	

Significant codes: '***' 0.001 '**' 0.01 '*' 0.5 '.' 0.1 ' ' 1

Table 1 presents the coefficients of the independent variables used in the linear regression model. We aimed to assess the impacts of various factors on food security. So, when all the predictor variables are zero, the food security value is computed to be 0.4199. The regression results based on the HRVS dataset alone (N=6000) indicate that earthquake exposure has no effect on food security. we found no evidence that households that experienced greater earthquake shaking intensity were significantly more likely to be food insecure. According to Thorne-Lyman et al. (2018), relief and humanitarian help stabilized food security during the 2015 earthquake. Between 2014 and 2016, the study region's rate of food insecurity dropped significantly from 17.6% to 12.4%. Researchers made use of the Household Food Insecurity Access Scale, a set of questions that reflects various aspects of food security, including food access, and has been evaluated in numerous worldwide research.

In the results, the dataset suggested that if a family member is unable to work due to illness, it negatively impacts the food security of the household. Sickness, creating a shortage of agricultural labour for the household affects the home production, hence, negatively impacts the food security of the household (Tuladhar, Sapkota, Adhikari, 2014). Based on the regression analysis, it appears that government subsidies are really having a negative impact on food security. This shows that the allocation of government subsidies isn't as successful as it could be. One key reason for the government's seeming incapacity to provide very basic support for farmers is its brief tenure. The purchase of agriculture inputs is made possible by subsidies, either from the government or from donors; nonetheless, misuse and limited availability are significant drawbacks. Due to incorrect implementation methods, the money obtained from bilateral and multilateral organizations did not address sustainability challenges (Bhandari, 2023).

Years of education appear to be a statistically significant predictor and positively impact food security, according to the regression result. When people who are food insecure are given food literacy training, they can significantly alter how they use food (West, Lindberg, Ball, McNaughton, 2020). The availability and accessibility components of food security are addressed by road linkage. About half of the people in our sample were from hilly or mountainous areas. Nepal lacks enough food because of its small amount of cultivable land, its slow rate of expansion, and its frequent dry winters, which reduce agricultural yields. It is noted that, particularly in rural areas where there is a

food deficit, international aid and public intervention continue to be significant sources of food supply for Nepal. Consequently, the nation's total ability to address the needs of marginalized groups is constrained, particularly in regions where the private sector is not very present (WFP, FAO, 2007). In our analysis, health costs and overall income are considered statistically unimportant. On food security, however, home production value (in NRs) and home purchase value (in NRs) have the opposite effects. Food security is positively correlated with house production and negatively correlated with home purchases. The low purchasing power of Nepali rural households is the reason for the detrimental effect of home purchase value on food security (WFP, FAO, 2007). There is a negative correlation between household food security and earthquake loss (in NRs), which is considered statistically significant. The magnitude of the earthquake has no bearing on food security because food help is available; nonetheless, a household is more susceptible to food insecurity if substantial assets were lost in the earthquake (SAWTEE, 2015).

6. Conclusion

The study gives useful insights into the numerous elements influencing food security in post-disaster Nepal. It emphasizes how important it is to implement focused interventions in sectors like healthcare, education, and subsidy management in order to improve households' general well-being and food security. The results also highlight how crucial it is to take local context and difficulties into account when developing successful policies and interventions.

Previous research has shown that extreme weather events, such as heat waves, hurricanes, floods, and droughts, are occurring more frequently and with greater intensity. This trend is expected to continue in the future due to ongoing anthropogenic climate change (Stocker, Qin, Plattner, Alexander, Allen, Bindoff, Xie, 2013). As a result, communities will face more health and well-being issues as they are exposed to more complex occurrences. The prompt aid response to the 2015 earthquake was a major contributing factor to the stabilization of food security in the impacted areas. However, Nepal might not be able to secure the same degree of assistance for other compound effects, such as landslides, floods, and drought. A population that experiences more environmental shocks will be more susceptible to socioeconomic problems like food insecurity, undernutrition and poverty. To effectively handle such shocks, it is crucial to comprehend which factors behave in which ways in various contexts and to create a policy in an objective way.

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